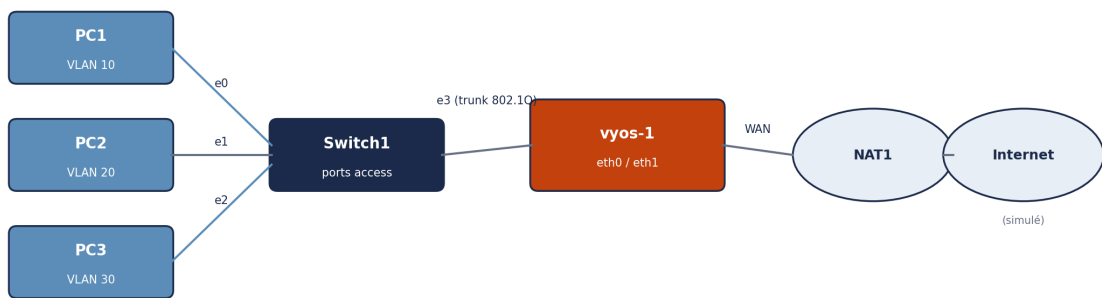


Lab VyOS — VLANs, DHCP & NAT

Topologie et configuration — GNS3

Routeur	VyOS 1.5-stream-2025-Q2
Plateforme	GNS3 2.2.55 / QEMU 6.2.0
Auteur	Nedia — Nettora

Topologie



Plan d'adressage : 192.168.10.0/24 (VLAN10) - 192.168.20.0/24 (VLAN20) - 192.168.30.0/24 (VLAN30)

Plan d'adressage

VLAN	Poste	Réseau	Passerelle	Plage DHCP
10	PC1	192.168.10.0/24	192.168.10.1	192.168.10.100 – .200
20	PC2	192.168.20.0/24	192.168.20.1	192.168.20.100 – .200
30	PC3	192.168.30.0/24	192.168.30.1	192.168.30.100 – .200

Configuration Switch1

Port	VLAN	Type	Connecté à
e0	10	access	PC1
e1	20	access	PC2
e2	30	access	PC3
e3	10, 20, 30	trunk (dot1q)	vynos-1 (eth0)

Configuration VyOS

Interfaces réseau

```
configure

set interfaces ethernet eth0 description LAN
set interfaces ethernet eth0 vif 10 address 192.168.10.1/24
set interfaces ethernet eth0 vif 10 description VLAN10-PC1
set interfaces ethernet eth0 vif 20 address 192.168.20.1/24
set interfaces ethernet eth0 vif 20 description VLAN20-PC2
set interfaces ethernet eth0 vif 30 address 192.168.30.1/24
set interfaces ethernet eth0 vif 30 description VLAN30-PC3

set interfaces ethernet eth1 address dhcp
set interfaces ethernet eth1 description WAN
```

Serveur DHCP

```
set service dhcp-server shared-network-name VLAN10 subnet 192.168.10.0/24 \
default-router 192.168.10.1
set service dhcp-server shared-network-name VLAN10 subnet 192.168.10.0/24 \
name-server 8.8.8.8
set service dhcp-server shared-network-name VLAN10 subnet 192.168.10.0/24 \
range 0 start 192.168.10.100
set service dhcp-server shared-network-name VLAN10 subnet 192.168.10.0/24 \
range 0 stop 192.168.10.200

(répéter pour VLAN20 -> 192.168.20.0/24 et VLAN30 -> 192.168.30.0/24)
```

NAT (translation vers Internet)

```
set nat source rule 100 outbound-interface name eth1
set nat source rule 100 source address 192.168.10.0/24
set nat source rule 100 translation address masquerade

set nat source rule 200 outbound-interface name eth1
set nat source rule 200 source address 192.168.20.0/24
set nat source rule 200 translation address masquerade

set nat source rule 300 outbound-interface name eth1
set nat source rule 300 source address 192.168.30.0/24
set nat source rule 300 translation address masquerade
```

Application

```
commit
save
```

Test (sur chaque PC VPCS)

```
dhcp
ping 192.168.10.1
ping 8.8.8.8
```

Poste	VLAN	DHCP	Ping passerelle	Ping Internet
PC1	10	OK	OK	OK
PC2	20	OK	OK	OK
PC3	30	OK	OK	OK